



Staurolite

△ **Staurolite in schist** from Russia showing cross-shaped twinning

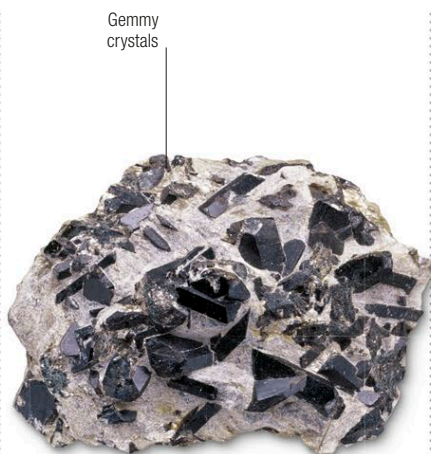
Staurolite is hydrous iron magnesium aluminium silicate. It occurs with garnet, tourmaline, and kyanite or sillimanite in mica schists and gneisses and other metamorphosed aluminium-rich rocks. Staurolite is reddish-brown or yellowish-brown, or nearly black, and normally occurs as prisms which are hexagonal or diamond-shaped in section. Staurolite is named from the Greek *stauros*, “cross”, and *lithos*, “stone”, for its cross-like twinned form. These cross-shaped crystals are frequently set in silver for use in religious jewellery.

Specification

Chemical name Aluminium silicate | **Formula**
 $(\text{Fe}, \text{Mg})_4\text{Al}_{17}(\text{Si}, \text{Al})_8\text{O}_{45}(\text{OH})_3$ | **Colour** Brown | **Structure**
 Monoclinic | **Hardness** 7–7.5 | **SG** 3.7 | **RI** 1.74–1.75
Lustre Vitreous to resinous | **Streak** Colourless to grey
Locations USA, France, Brazil



Staurolite and kyanite specimen | Rough | Staurolite and kyanite often occur together, as in this specimen of matrix of muscovite mica schist.



Staurolite schist | Rough | Staurolite commonly occurs in a groundmass of mica schist. In this example, the crystals are rich brown and gemmy.



Carved sphere | Carved | Small crystals are set in a feldspar and mica matrix in this rare sphere. The material originates from the Kola Peninsula of Russia.



Twin staurolite crystal | Rough | This specimen of cross-shaped crystal of staurolite displays typically geometric lines caused by twinning.



Single crystal | Rough | This twin of staurolite has been separated from its matrix. These crystals are frequently mounted as pendants.